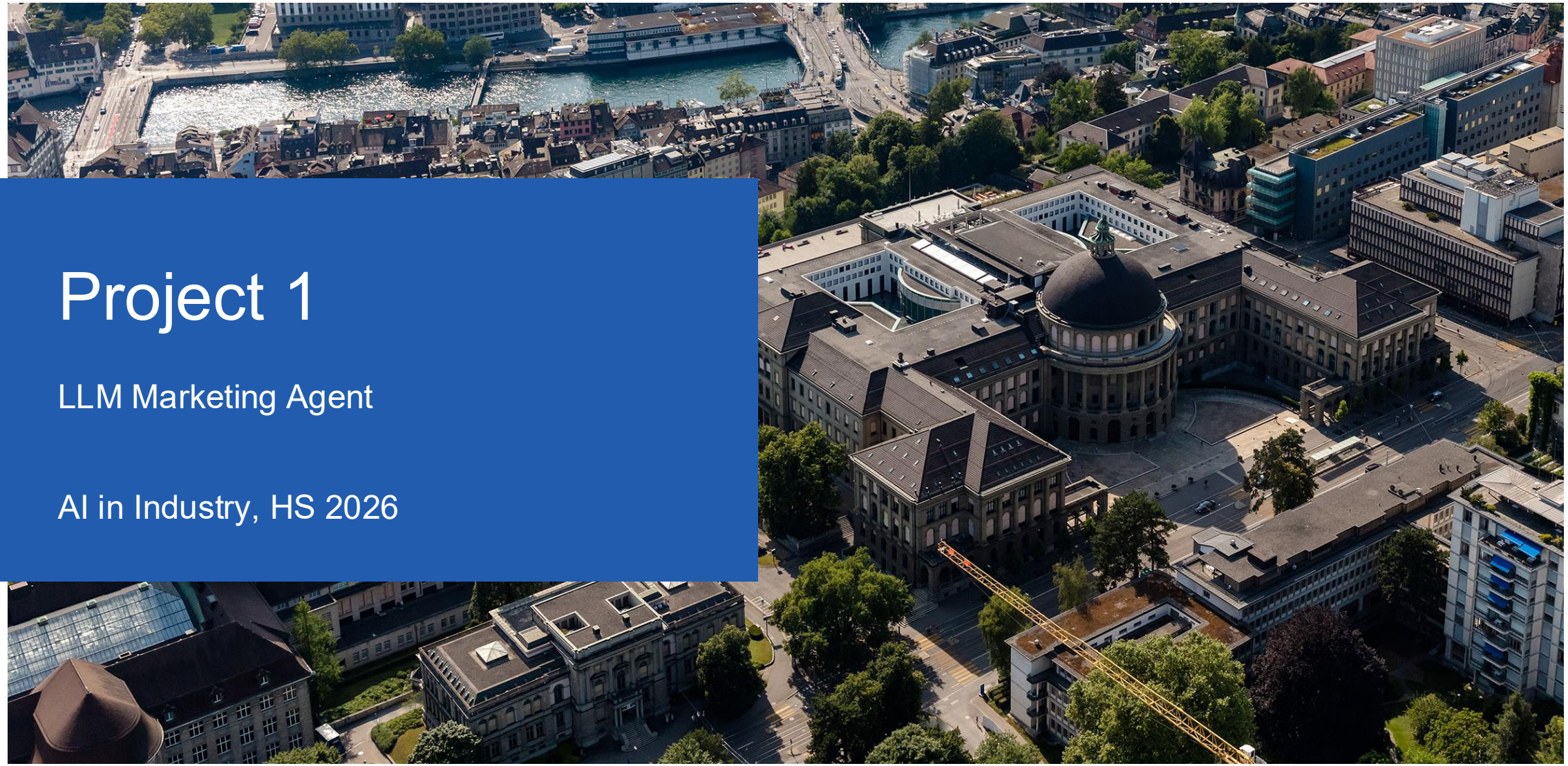
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Project 1

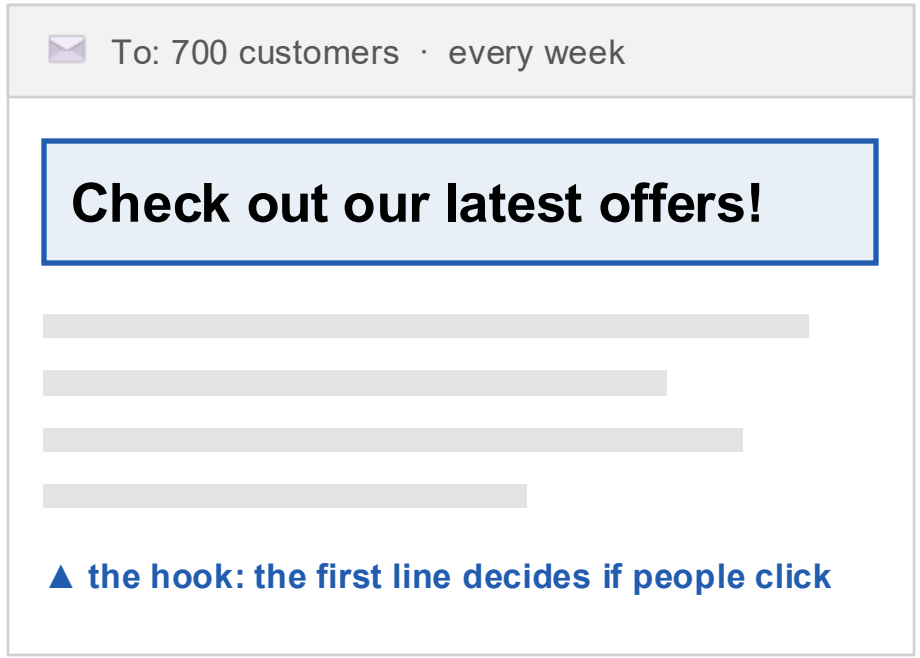
LLM Marketing Agent

AI in Industry, HS 2026

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Motivation

Project: AI marketing agent



- You work in the marketing team of an **online clothing shop** in Switzerland with **700 customers**.
- Every week the shop sends an email. Whether a customer visits the shop depends mostly on its first line: the **hook**.
- You build an **AI marketing agent** to generate hooks that ***maximize the number of customer visits***.
- This is done with OPRO: You steer the agent with **instructions in plain English (meta-prompt)**

Challenges



Different customers, different wishes

A new customer, a big spender and an unhappy customer do not react to the same email.



You cannot read their minds

You only see how many customers visited the shop, plus a little customer feedback.



No formula for a good email

"Good" is a matter of language and taste, not something you can compute.

Idea: let a language model try hooks, show it how each one did, and ask it for better ones. Repeat.

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The World

Simulation of real-world customer behavior

What is in the World?



Customers

700 customers of the shop, with age, number of orders, order value, days since last order and satisfaction.



Segments

Five groups that want different things.

N H R L A



Hidden customers

For each segment, a language model plays a customer with a secret description.



Marketing agent

Writes one hook per segment and learns from the results. **You build this.**

Everything is **simulated**, so you can run a whole marketing season in a few minutes.

Five customer segments

Every customer is one **dictionary** in Python:

```
{
    'name': 'Marco Roth',
    'age': 39,
    'n_purchases': 4,
    'avg_order_value': 28.7,
    'days_since_last_purchase': 53,
    'satisfaction': 0.71
}
```

N **New**

Signed up recently but have not bought anything yet.

130 customers

H **High-value**

Buy often and spend a lot per order.

130 customers

R **Regular**

Buy from us now and then.

150 customers

L **Low-budget**

Buy cheap items and spend little per order.

150 customers

A **At-risk**

Used to buy regularly; gone quiet or unhappy.

140 customers

Simulation: hidden persona for each customer segment



The shop's current hook, scored by each hidden customer:

N **0.19** H **0.23** R **0.28** L **0.20** A **0.27**

For you it is a **black box**: text goes in, a number comes out. The score says how strongly the customer leans towards YES.

What you see, and what stays hidden

👀 You see

- The **customer data**: 700 dictionaries
- A short **description** of each segment, e.g. *"Buy cheap items and spend little per order."*
- **Customer feedback** from surveys and support tickets, e.g. *"Shipping costs make small orders not worth it."*
- The **visit rate** of every campaign you send.

🔒 Hidden

- What each segment's customer **really wants**, and what puts them off.
- Which hook would be the best one.

The marketing campaign cycle

1

The shop sends a campaign

Every customer gets the hook of their segment.

✉ **Hook per segment**

→

2

The hidden customer reads it

The language model gives the hook a score between 0 and 1.

🎭 **Score**

→

3

Each customer flips a biased coin

A better score means a better chance to visit, but no guarantee.

🎲 **Visit probability**

→

4

Count the visits

The share of each segment that visited: the number the agent tries to push up.

📊 **Visit rate**

visit probability = 5% + 60% × score

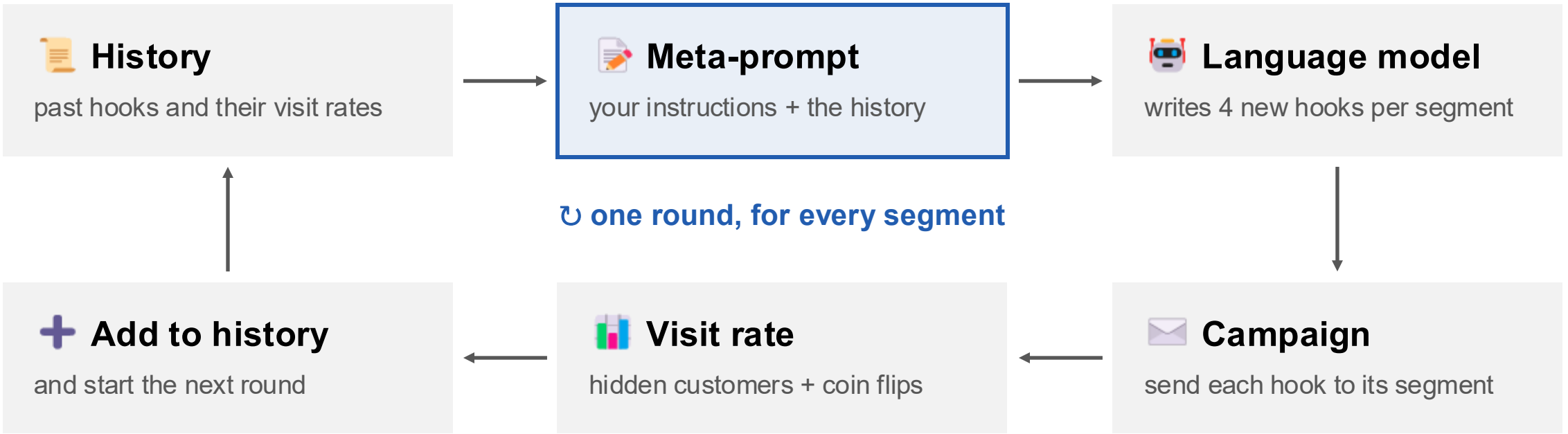
A hook with score 0.5 gives each customer a 5% + 60% × 0.5 = **35%** chance to visit. Even a perfect hook (score 1) brings at most **65%**.

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The agent: OPRO

Optimization by PROmpting

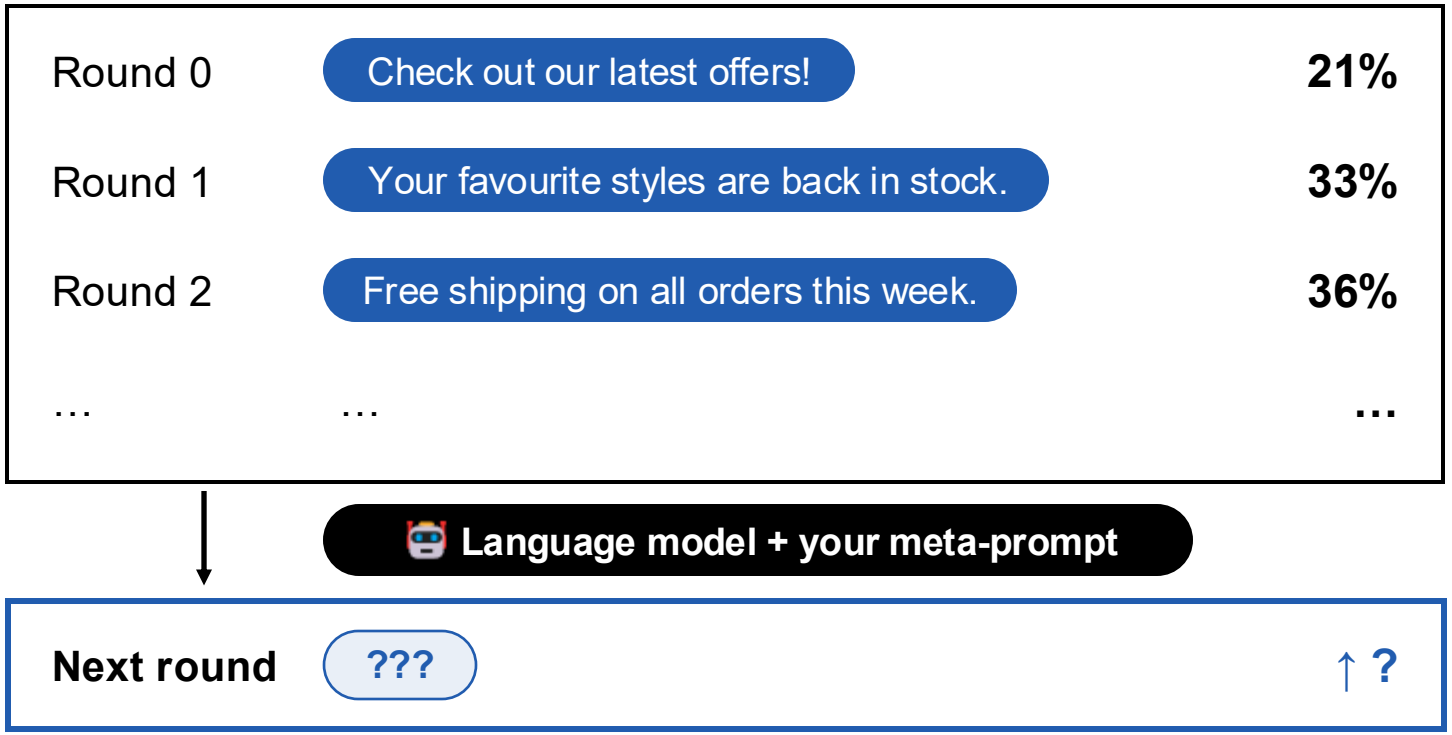
OPRO: Optimization by PROmpting (ICLR'24, arXiv:2309.03409)



Goal: step by step, discover the hooks that make the most customers of each segment visit the shop.

OPRO, round by round

History of one segment, sorted from worst to best (the best one last)



- Each round the model reads the **k best** past hooks with their visit rates.
- It writes **4 new hooks** per segment; all of them are sent and measured.
- **6 rounds**, all five segments at once.

The meta-prompt: your main lever 🎯 TODO 7

You start from this (it works, but barely):

```
Write a marketing email hook.

Past hooks:
{past_hooks}

Reply with the hook only.
```

There is no single right answer. Different meta-prompts give different results, and you explain your choices in the defence.

Ideas to make it better:

- Give the model a **role** and explain the situation.
- Tell it **who the customers are** (the segment description).
- Pass on the **customer feedback**: the only real clue.
- Show the **past hooks** and say what the numbers mean.
- Ask it to **think first**: what do these customers care about, what stops them?
- Ask for something **new and concrete**, not a small variation.
- Set **rules**: length, no placeholders like [Name].

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Tasks

What you implement in the notebook

TODOs: 9 tasks



PART 1
Meet your customers
TODO 1
Write hooks by hand and see how customers react.

PART 2
Segments
TODO 2–3
Sort customers with five rules and count them.

PART 3
Campaign
TODO 4–5
Visit probability and visit rate per segment.

PART 4
OPRO agent
TODO 6–7
Format the history and write the meta-prompt.

PART 5
Run & experiment
TODO 8–9
Compare with the baseline, then iterate your meta-prompt design.

- **Basic Python implementations** with hints
- After most TODOs, a **check cell** tells you if it works.

```
✅ score 0 gives just the base rate: expected 0.05, got 0.05
✅ score 1 gives base rate + lift: expected 0.65, got 0.65

All checks passed. 🎉
```